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Coating film, tool, and machine tool

Abstract

A coating film has a lamination unit including a first layer and at least one of a second layer and a third layer. The first layer is a nitride or the like of a first material represented by (Cr.sub.1-a-b-cAl.sub.a[Ni.sub.1-dZr.sub.d].sub.bX.sub.c). X is at least one element selected from Ti, Nb, Si, B, W, and V. a, b, c, and d represent atomic concentrations. The second layer is a nitride or the like of the second material represented by (Al.sub.eCr.sub.1-e-fZ.sub.f). Z is at least one element selected from Si, Y, and B. e and f represent atomic concentrations. The third layer is a nitride or the like of the third material represented by (Al.sub.gCr.sub.1-g).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present invention claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2022-031561 filed on Mar. 2, 2022, the entire content of which is incorporated herein by reference.

FIELD OF THE INVENTION

(2) The present disclosure relates to a coating film, a tool, and a machine tool.

BACKGROUND

(3) Conventionally, a machine tool that cuts a part such as a gear is known. A machine tool cuts a part by bringing a tool such as a hob cutter, a pinion cutter, a skiving cutter, or a shaving cutter into contact with a surface of the part. This type of tool has a base material and a coating film covering the surface of the base material. For example, an Al—Cr—N-based hard coating film is used for

the coating film.

(4) Since the cutting tool cuts a metal part, the coating film on the surface is required to have very high hardness. However, since a material having high hardness has low toughness, there is a problem that cracks and wear are likely to occur. For this reason, it has been conventionally difficult to use a material having high hardness and to realize a coating film having high abrasion resistance.

SUMMARY

(5) A coating film according to an exemplary embodiment of the present disclosure includes a lamination unit including a first layer and at least one of a second layer and a third layer. The first layer contains a nitride, a carbide, a carbonitride, or an oxide of a first material of which elemental composition is represented by $(\text{Cr}_{1-a-b-c}\text{Al}_a[\text{Ni}_{1-d}\text{Zr}_d]_b\text{X}_c)$. The $[\text{Ni}_{1-d}\text{Zr}_d]$ is a ZrNi compound. The X is at least one element selected from Ti, Nb, Si, B, W, and V. The a, b, and c respectively represent atomic concentrations of Al, [NiZr], and X in the $(\text{Cr}_{1-a-b-c}\text{Al}_a[\text{Ni}_{1-d}\text{Zr}_d]_b\text{X}_c)$, and satisfy $0.5 \leq a \leq 0.8$, $0.01 \leq b \leq 0.35$, and $0 \leq c \leq 0.2$. The d represents an atomic concentration of Zr in the $[\text{Ni}_{1-d}\text{Zr}_d]$, and satisfies $0.2 \leq d \leq 0.5$. The second layer contains a nitride, a carbide, a carbonitride, or an oxide of a second material of which elemental composition is represented by $(\text{Al}_c\text{Cr}_{1-e-f}\text{Z}_f)$. The Z is at least one element selected from Si, Y, and B. The e and f represent atom concentrations of Al and Z in the $(\text{Al}_c\text{Cr}_{1-e-f}\text{Z}_f)$, respectively, and satisfy $0.5 \leq e \leq 0.8$, $0.03 \leq f \leq 0.3$, $e+f \leq 0.9$. The third layer contains a nitride, a carbide, a carbonitride, or an oxide of a third material of which elemental composition is represented by $(\text{Al}_g\text{Cr}_{1-g})$. The g represents an atomic concentration of Al in the $(\text{Al}_g\text{Cr}_{1-g})$, and satisfies $0.5 \leq g \leq 0.8$.

(6) The above and other elements, features, steps, characteristics and advantages of the present disclosure will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side view of a machine tool;
- (2) FIG. 2 is a plan view of the machine tool;
- (3) FIG. 3 is a perspective view of a tool;
- (4) FIG. 4 is a cross-sectional view of the vicinity of a cutting edge of the tool;
- (5) FIG. 5 is a diagram schematically illustrating a layer structure of a coating film according to a first embodiment;
- (6) FIG. 6 is a diagram schematically illustrating a layer structure of a coating film according to a second embodiment;
- (7) FIG. 7 is a diagram schematically illustrating a layer structure of a coating film according to a third embodiment;
- (8) FIG. 8 is a diagram schematically illustrating a layer structure of a coating film according to a fourth embodiment;
- (9) FIG. 9 is a graph illustrating results of an experiment for examining abrasion resistance performance of a coating film; and
- (10) FIG. 10 is a perspective view of the tool.

DETAILED DESCRIPTION

(11) FIG. 1 is a side view of a machine tool 1 according to an embodiment. FIG. 2 is a plan view of the machine tool 1. FIGS. 1 and 2 illustrate an x direction, a y direction, and a z direction. In the present embodiment, the x direction and the y direction are horizontal directions and directions orthogonal to each other. In the present embodiment, the z direction is a vertical direction.

However, the definitions of the x direction, the y direction, and the z direction do not limit the posture of the machine tool **1**. That is, the x direction and the y direction may be directions other than the horizontal direction. Further, the z direction may be a direction other than the vertical direction.

(12) The machine tool **1** is a device that manufactures a gear by cutting a metal part W with a tool **30**. The machine tool **1** of the present embodiment is a so-called hob board using a hob cutter as the tool **30**. As illustrated in FIG. **1**, the machine tool **1** includes a bed **10**, a tool holder **20**, a tool **30**, a part holder **40**, and a part changer **50**.

(13) The bed **10** is a support base that supports the tool holder **20** and the part holder **40**. The bed **10** is formed of rolled steel having high rigidity. The bed **10** is installed on a floor surface of a factory that manufactures gears. In FIG. **1**, the bed **10** has a flat plate shape, and the tool holder **20** and the part holder **40** are supported on an upper surface of the bed. However, the bed **10** may have a housing shape having a bottom plate portion, a side wall portion, and a top plate portion, and the tool holder **20** and the part holder **40** may be accommodated in the housing shape. The tool holder **20** and the part holder **40** are arranged side by side in the x direction.

(14) The tool holder **20** is a mechanism that holds the tool **30**. As illustrated in FIGS. **1** and **2**, the tool holder **20** includes a column **21**, a saddle **22**, and a tool head **23**.

(15) The column **21** is movable in the x direction with respect to the bed **10** by a first drive mechanism (not illustrated). The saddle **22** is provided on a surface of the column **21** facing the part holder **40**. The saddle **22** is movable in the z direction with respect to the column **21** by a second drive mechanism (not illustrated).

(16) The tool head **23** is attached to the saddle **22**. The tool head **23** is movable in the y direction with respect to the saddle **22** by a third drive mechanism (not illustrated). The tool head **23** is rotatable about a rotation axis **9** extending in the y direction by a fourth drive mechanism (not illustrated).

(17) The tool **30** is a component for machining the part W. The tool **30** is detachable from the tool head **23**. When the first drive mechanism, the second drive mechanism, and the third drive mechanism described above are operated with the tool **30** attached to the tool head **23**, the tool **30** moves in the x direction, the y direction, and the z direction together with the tool head **23**. When the above-described fourth drive mechanism is operated with the tool **30** attached to the tool head **23**, the tool **30** rotates at a high speed about the rotation axis **9** extending in the y direction.

(18) FIG. **3** is a perspective view of the tool **30**. The tool **30** of the present embodiment is a hob cutter for cutting external teeth of a gear. The tool **30** has a substantially cylindrical shape, and has a large number of cutting blades **31** arranged in a spiral shape on an outer peripheral surface thereof. In a state where the tool **30** is attached to the tool head **23**, the rotation axis **9** of the tool head **23** and the central axis of the tool **30** are coaxially arranged.

(19) The part holder **40** is a mechanism that supports the part W, which is a part. The part holder **40** is disposed spaced apart from the tool holder **20** in the x direction. As illustrated in FIG. **1**, the part holder **40** includes a table **41**, a counter column **42**, and a tailstock **43**. In FIG. **2**, illustration of the counter column **42** and the tailstock **43** is omitted.

(20) The table **41** is a unit that supports the part W from below. The table **41** is disposed between the column **21** and the counter column **42**. A lower damper **411** that supports the part W from below is provided at the upper end of the table **41**.

(21) The counter column **42** is fixed to the upper surface of the bed **10**. The counter column **42** extends upward from the upper surface of the bed **10**. The tailstock **43** is a unit that supports the part W from above. The tailstock **43** is provided on a surface of the counter column **42** facing the tool holder **20** so as to be vertically movable. An upper damper **431** that supports the part W from above is provided at a lower end of the tailstock **43**.

(22) The part W is held in a sandwiched state between the lower damper **411** of the table **41** and the upper damper **431** of the tailstock **43**. The lower damper **411** and the upper damper **431** are

rotatable about a rotation axis extending in the z direction by a rotation mechanism (not illustrated). Therefore, the part W can be rotated by operating the rotation mechanism in a state where the part W is held by the lower clasper **411** and the upper damper **431**.

(23) The part changer **50** is a mechanism that conveys the part W. As illustrated in FIGS. **1** and **2**, the part changer **50** includes a turntable **51**. The turntable **51** has a plurality of arms **52**. The part W is held at the distal end of each arm **52** via a gripper. The turntable **51** rotates about a rotation axis extending in the z direction by a rotation mechanism (not illustrated) while holding the part W in each arm **52**. As a result, the part W can be conveyed between a machining position P1 between the table **41** and the tailstock **43** and a standby position P2 laterally away from the machining position P1.

(24) In the example of FIG. **2**, the turntable **51** includes three arms **52**. However, the number of arms **52** of the turntable **51** may be 1, 2, or 4 or more.

(25) When the part W is machined by the machine tool **1**, first, the part changer **50** brings the part W into the machining position P1 from the standby position P2. Then, the part W is held by the lower clasper **411** and the upper damper **431**. Next, the rotation of the part W about the rotation axis extending in the z direction is started by the above-described rotation mechanism, and the rotation of the tool **30** about the rotation axis extending in the y direction is started by the above-described fourth drive mechanism.

(26) Subsequently, the tool **30** is brought close to the part W by the above-described first drive mechanism. Then, while rotating the part W and the tool **30**, the position of the tool **30** with respect to the part W is gradually changed. As a result, external teeth are formed on the outer peripheral surface of the part W. Thereafter, the holding of the part W by the lower clasper **411** and the upper damper **431** is released, and the part W is unloaded from the machining position P1 to the standby position P2 by the part changer **50**.

(27) As described above, the machine tool **1** includes the part holder **40** and the tool **30** that machines the part W held by the part holder **40**. As described later, the tool **30** has a coating film **33** having high abrasion resistance. Therefore, the machine tool **1** can accurately machine the part W over a long period of time using the tool **30**.

(28) FIG. **4** is a cross-sectional view of the vicinity of the cutting blade **31** of the tool **30**. As illustrated in FIG. **4**, the tool **30** includes a base material **32** formed of an alloy such as high speed steel or cemented carbide, and a hard coating film **33** covering the surface of the base material **32**. Hereinafter, a first embodiment, a second embodiment, a third embodiment, and a fourth embodiment of the coating film **33** will be described.

(29) FIG. **5** is a diagram schematically illustrating a layer structure of the coating film **33** according to the first embodiment. As illustrated in FIG. **5**, the coating film **33** of the first embodiment is a multilayer film in which a plurality of lamination units **61** each including as a first layer L1 and a second layer L2 are stacked. The thickness of the coating film **33** is, for example, about 1 μm to 5 μm . As described later, by repeatedly laminating the lamination unit **61** a plurality of times, the abrasion resistance of the coating film **33** can be improved as compared with the case where the lamination is not repeated.

(30) The first layer L1 contains a nitride, a carbide, a carbonitride, or an oxide of a first material having an elemental composition represented by $(\text{Cr}_{\text{sub.1-a-b-c}}\text{Al}_{\text{sub.a}}[\text{Ni}_{\text{sub.1-dZr}_{\text{sub.d}}}]_{\text{sub.bX}_{\text{sub.c}}})$. In the above composition formula, Cr is chromium, Al is aluminum, Ni is nickel, and Zr is zirconium. $[\text{Ni}_{\text{sub.1-dZr}_{\text{sub.d}}}]$ is a ZrNi compound. X is at least one element selected from Ti (titanium), Nb (niobium), Si (silicon), B (boron), W (tungsten), and V (vanadium).

(31) a, b, and c represent the atomic concentrations (at %) of Al, $[\text{NiZr}]$, and X in $(\text{Cr}_{\text{sub.1-a-b-c}}\text{Al}_{\text{sub.a}}[\text{Ni}_{\text{sub.1-dZr}_{\text{sub.d}}}]_{\text{sub.bX}_{\text{sub.c}}})$, respectively. a satisfies $0.5 \leq a \leq 0.8$. b satisfies $0.01 \leq b \leq 0.35$. c satisfies $0 \leq c \leq 0.2$. d represents the atomic concentration of Zr in $[\text{Ni}_{\text{sub.1-dZr}_{\text{sub.d}}}]$. d satisfies $0.2 \leq d \leq 0.5$. By setting the atomic concentrations of Al, $[\text{NiZr}]$, and X within such numerical ranges, the first layer L1 can be a layer having high abrasion resistance. It is more

desirable that a , b , and c satisfy $0.51 \leq a+b+c < 1$.

(32) It is desirable that the first layer **L1** be substantially composed only of a nitride, a carbide, a carbonitride, or an oxide of the first material. However, the first layer **L1** may contain a nitride, a carbide, a carbonitride, or an oxide of the first material and impurities inevitably mixed.

(33) The second layer **L2** contains a nitride, a carbide, a carbonitride, or an oxide of the second material having an elemental composition represented by $(Al_{sub.c}Cr_{sub.1-e-f}Z_{sub.f})$. In the above composition formula, Al is aluminum, and Cr is chromium. Z is at least one element selected from Si (silicon), Y (yttrium), and B (boron). e and f represent the atomic concentrations (at %) of Al and Z in $(Al_{sub.c}Cr_{sub.1-e-f}Z_{sub.f})$, respectively. e satisfies $0.5 \leq e < 0.8$. f satisfies $0.03 \leq f < 0.3$. e and f satisfy $e+f \leq 0.9$. By setting the atomic concentrations of Al and Z within such a numerical range, the second layer **L2** can be a layer having high abrasion resistance. As described above, Si (silicon) or Y (yttrium) can be selected as Z . By selecting Si (silicon) or Y (yttrium), the abrasion resistance of the coating film **33** including the second layer **L2** can be improved.

(34) The second layer **L2** is desirably substantially composed only of a nitride, a carbide, a carbonitride, or an oxide of the second material. However, the second layer **L2** may contain a nitride, a carbide, a carbonitride, or an oxide of the second material and impurities inevitably mixed.

(35) The coating film **33** of the first embodiment is formed by alternately and repeatedly forming the first layer **L1** and the second layer **L2** on the surface of the base material **32**. As a film forming method, for example, an ion plating method which is a type of physical vapor deposition (PVD) is used. Specifically, in the chamber, the base material **32** is disposed at the center, and a first target formed of the first material and a second target formed of the second material are disposed around the base material. Then, vapor deposition of the first material from the first target and vapor deposition of the second material from the second target are performed while rotating the base material **32**. As a result, the first layer **L1** and the second layer **L2** are alternately and repeatedly deposited on the surface of the base material **32**.

(36) At this time, when nitrogen (N_2) gas as a nitriding source is introduced into the chamber, the first material and the second material become nitrides and are deposited on the surface of the base material **32**. When methane (CH_4) gas as a carbonization source is introduced into the chamber, the first material and the second material become carbides, and are deposited on the surface of the base material **32**. In addition, when nitrogen (N_2) gas as a nitriding source and methane (CH_4) gas as a carbonization source are introduced into the chamber, the first material and the second material become carbonitrides, and are deposited on the surface of the base material **32**. When oxygen (O_2) gas as an oxidation source is introduced into the chamber, the first material and the second material become oxides, and are deposited on the surface of the base material **32**.

(37) As described above, both the first layer **L1** and the second layer **L2** have excellent abrasion resistance. However, the hardness of the first layer **L1** is relatively lower than that of the second layer **L2**. By sandwiching the first layer **L1** between the second layers **L2**, a buffering action by the first layer **L1** is obtained, and the toughness of the coating film **33** is improved. Therefore, minute cracks are less likely to occur on the surface of the coating film **33** than in a case where the coating film includes only the first layer **L1** or only the second layer **L2**. As a result, the abrasion resistance of the coating film **33** is improved, and the life of the tool **30** can be prolonged.

(38) The thickness h of the lamination unit **61** constituted by the first layer **L1** and the second layer **L2** is desirably 200 nm or less. By suppressing the thickness h of the lamination unit **61** to 200 nm or less, the lamination unit **61** can be repeatedly stacked in the coating film **33** many times. Accordingly, the abrasion resistance of the coating film **33** can be further improved.

(39) In the example of FIG. 5, the lowermost layer of the coating film **33** is the first layer **L1**, and the uppermost layer is the second layer **L2**. However, the lowermost layer of the coating film **33** may be the second layer **L2**. The uppermost layer of the coating film **33** may be the first layer **L1**.

(40) FIG. 6 is a diagram schematically illustrating a layer structure of the coating film **33** according

to the second embodiment. As illustrated in FIG. 6, the coating film 33 of the second embodiment is a multilayer film in which a plurality of lamination units 62 constituted by the first layer L1 and the third layer L3 are stacked. The thickness of the coating film 33 is, for example, about 1 μm to 5 μm . As described later, by repeatedly laminating the lamination unit 62 a plurality of times, the abrasion resistance of the coating film 33 can be improved as compared with the case where the lamination is not repeated.

(41) The first layer L1 contains a nitride, a carbide, a carbonitride, or an oxide of the first material having the same elemental composition as that of the first embodiment described above. It is desirable that the first layer L1 be substantially composed only of a nitride, a carbide, a carbonitride, or an oxide of the first material. However, the first layer L1 may contain a nitride, a carbide, a carbonitride, or an oxide of the first material and impurities inevitably mixed.

(42) The third layer L3 includes a nitride, a carbide, a carbonitride, or an oxide of the third material having an elemental composition represented by $(\text{Al.sub.gCr.sub.1-g})$. In the above composition formula, Al is aluminum, and Cr is chromium. g represents the atomic concentration (at %) of Al in $(\text{Al.sub.gCr.sub.1-g})$. g satisfies $0.5 \leq g \leq 0.8$. By setting the atomic concentration of Al within such a numerical range, the third layer L3 can be a layer having high abrasion resistance.

(43) It is desirable that the third layer L3 be substantially made only of a nitride, a carbide, a carbonitride, or an oxide of the third material. However, the third layer L3 may contain a nitride, a carbide, a carbonitride, or an oxide of the third material and impurities inevitably mixed.

(44) The coating film 33 of the second embodiment is formed by alternately and repeatedly forming the first layer L1 and the third layer L3 on the surface of the base material 32. As a film forming method, for example, an ion plating method which is a type of physical vapor deposition (PVD) is used. Specifically, in the chamber, the base material 32 is disposed at the center, and the first target formed of the first material and the third target formed of the third material are disposed around the base material. Then, vapor deposition of the first material from the first target and vapor deposition of the third material from the third target are performed while rotating the base material 32. As a result, the first layer L1 and the third layer L3 are alternately and repeatedly deposited on the surface of the base material 32.

(45) At this time, when nitrogen (N_2) gas as a nitriding source is introduced into the chamber, the first material and the third material become nitrides and are deposited on the surface of the base material 32. When a methane (CH_4) gas as a carbonization source is introduced into the chamber, the first material and the third material become carbides, and are deposited on the surface of the base material 32. In addition, when nitrogen (N_2) gas as a nitriding source and methane (CH_4) gas as a carbonization source are introduced into the chamber, the first material and the third material become carbonitrides, and are deposited on the surface of the base material 32. When oxygen (O_2) gas as an oxidation source is introduced into the chamber, the first material and the third material become oxides, and are deposited on the surface of the base material 32.

(46) As described above, both the first layer L1 and the third layer L3 have excellent abrasion resistance. However, the hardness of the first layer L1 is relatively lower than that of the third layer L3. By sandwiching the first layer L1 between the third layers L3, a buffering action by the first layer L1 is obtained, and the toughness of the coating film 33 is improved. Therefore, minute cracks are less likely to occur on the surface of the coating film 33 than in a case where the coating film includes only the first layer L1 or only the third layer L3. As a result, the abrasion resistance of the coating film 33 is improved, and the life of the tool 30 can be prolonged.

(47) The thickness h of the lamination unit 62 constituted by the first layer L1 and the third layer L3 is desirably 200 nm or less. By suppressing the thickness h of the lamination unit 62 to 200 nm or less, the lamination unit 62 can be repeatedly stacked in the coating film 33 many times. Accordingly, the abrasion resistance of the coating film 33 can be further improved.

(48) In the example of FIG. 6, the lowermost layer of the coating film 33 is the first layer L1, and the uppermost layer is the third layer L3. However, the lowermost layer of the coating film 33 may

be the third layer L3. The uppermost layer of the coating film 33 may be the first layer L1.

(49) FIG. 7 is a diagram schematically illustrating a layer structure of the coating film 33 according to the third embodiment. As illustrated in FIG. 7, the coating film 33 of the third embodiment is a multilayer film in which a plurality of lamination units 63 including a first layer L1, a second layer L2, and a third layer L3 are stacked. The thickness of the coating film 33 is, for example, about 1 μm to 5 μm . As described later, by repeatedly laminating the lamination unit 63 a plurality of times, the abrasion resistance of the coating film 33 can be improved as compared with the case where the lamination is not repeated.

(50) The first layer L1 contains a nitride, a carbide, a carbonitride, or an oxide of the first material having the same elemental composition as that of the first embodiment described above. It is desirable that the first layer L1 be substantially composed only of a nitride, a carbide, a carbonitride, or an oxide of the first material. However, the first layer L1 may contain a nitride, a carbide, a carbonitride, or an oxide of the first material and impurities inevitably mixed.

(51) The second layer L2 contains a nitride, a carbide, a carbonitride, or an oxide of the second material having the same elemental composition as that of the first embodiment described above. The second layer L2 is desirably substantially composed only of a nitride, a carbide, a carbonitride, or an oxide of the second material. However, the second layer L2 may contain a nitride, a carbide, a carbonitride, or an oxide of the second material and impurities inevitably mixed.

(52) The third layer L3 includes a nitride, a carbide, a carbonitride, or an oxide of the third material having the same elemental composition as that of the second embodiment described above. It is desirable that the third layer L3 be substantially made only of a nitride, a carbide, a carbonitride, or an oxide of the third material. However, the third layer L3 may contain a nitride, a carbide, a carbonitride, or an oxide of the third material and impurities inevitably mixed.

(53) The coating film 33 of the third embodiment is formed by repeatedly forming the first layer L1, the second layer L2, and the third layer L3 on the surface of the base material 32. As a film forming method, for example, an ion plating method which is a type of physical vapor deposition (PVD) is used. Specifically, in the chamber, the base material 32 is disposed at the center, and the first target formed of the first material, the second target formed of the second material, and the third target formed of the third material are disposed around the base material. Then, while rotating the base material 32, vapor deposition of the first material from the first target, vapor deposition of the second material from the second target, and vapor deposition of the third material from the third target are performed. As a result, the first layer L1, the second layer L2, and the third layer L3 are repeatedly formed on the surface of the base material 32.

(54) At this time, when nitrogen (N_2) gas as a nitriding source is introduced into the chamber, the first material, the second material, and the third material become nitrides and are deposited on the surface of the base material 32. When methane (CH_4) gas as a carbonization source is introduced into the chamber, the first material, the second material, and the third material become carbides, and are deposited on the surface of the base material 32. When nitrogen (N_2) gas as a nitriding source and methane (CH_4) gas as a carbonization source are introduced into the chamber, the first material, the second material, and the third material become carbonitrides, and are deposited on the surface of the base material 32. When oxygen (O_2) gas as an oxidation source is introduced into the chamber, the first material, the second material, and the third material become oxides, and are deposited on the surface of the base material 32.

(55) As described above, each of the first layer L1, the second layer L2, and the third layer L3 has excellent abrasion resistance. However, the first layer L1 has relatively lower hardness than the second layer L2 and the third layer L3. By sandwiching such a first layer L1 between the second layer L2 and the third layer L3, a buffering action by the first layer L1 is obtained, and the toughness of the coating film 33 is improved. Therefore, minute cracks are less likely to occur on the surface of the coating film 33 than in a case where the coating film includes only the first layer L1, only the second layer L2, or only the third layer L3. As a result, the abrasion resistance of the

coating film **33** is improved, and the life of the tool **30** can be prolonged.

(56) The thickness h of the lamination unit **63** including the first layer **L1**, the second layer **L2**, and the third layer **L3** is desirably 200 nm or less. By suppressing the thickness h of the lamination unit **63** to 200 nm or less, the lamination unit **63** can be repeatedly stacked in the coating film **33** many times. Accordingly, the abrasion resistance of the coating film **33** can be further improved.

(57) In the example of FIG. 7, the lowermost layer of the coating film **33** is the first layer **L1**, and the uppermost layer is the third layer **L3**. However, the lowermost layer of the coating film **33** may be the second layer **L2** or the third layer **L3**. The uppermost layer of the coating film **33** may be the first layer **L1** or the second layer **L2**.

(58) In addition, the stacking order of the first layer **L1**, the second layer **L2**, and the third layer **L3** is not limited to the example of FIG. 7. For example, the stacking order of the second layer **L2** and the third layer **L3** may be opposite to that in the example of FIG. 7.

(59) FIG. 8 is a diagram schematically illustrating a layer structure of the coating film **33** according to the fourth embodiment. As illustrated in FIG. 8, the coating film **33** of the fourth embodiment is a multilayer film having a lower layer portion **331** and an upper layer portion **332**. The thickness of the coating film **33** is, for example, about 1 μm to 5 μm . In the lower layer portion **331**, a plurality of lamination units **63** including the first layer **L1**, the second layer **L2**, and the third layer **L3** are stacked. In the upper layer portion **332**, a plurality of lamination units **61** including the first layer **L1** and the second layer **L2** are stacked. As described later, by repeatedly laminating the lamination unit **61** and the lamination unit **63** a plurality of times, the abrasion resistance of the coating film **33** can be improved as compared with the case where the lamination is not repeated.

(60) The first layer **L1** contains a nitride, a carbide, a carbonitride, or an oxide of the first material having the same elemental composition as that of the first embodiment described above. It is desirable that the first layer **L1** be substantially composed only of a nitride, a carbide, a carbonitride, or an oxide of the first material. However, the first layer **L1** may contain a nitride, a carbide, a carbonitride, or an oxide of the first material and impurities inevitably mixed.

(61) The second layer **L2** contains a nitride, a carbide, a carbonitride, or an oxide of the second material having the same elemental composition as that of the first embodiment described above. The second layer **L2** is desirably substantially composed only of a nitride, a carbide, a carbonitride, or an oxide of the second material. However, the second layer **L2** may contain a nitride, a carbide, a carbonitride, or an oxide of the second material and impurities inevitably mixed.

(62) The third layer **L3** includes a nitride, a carbide, a carbonitride, or an oxide of the third material having the same elemental composition as that of the second embodiment described above. It is desirable that the third layer **L3** be substantially made only of a nitride, a carbide, a carbonitride, or an oxide of the third material. However, the third layer **L3** may contain a nitride, a carbide, a carbonitride, or an oxide of the third material and impurities inevitably mixed.

(63) The coating film **33** of the fourth embodiment is formed by first forming the lower layer portion **331** on the surface of the base material **32** and then forming the upper layer portion **332**. At the time of film formation of the lower layer portion **331**, the first layer **L1**, the second layer **L2**, and the third layer **L3** are repeatedly formed. A specific film forming method is the same as that of the third embodiment described above. At the time of film formation of the upper layer portion **332**, the first layer **L1** and the second layer **L2** are repeatedly formed. A specific film forming method is the same as that of the first embodiment described above.

(64) When nitrogen (N_2) gas as a nitriding source is introduced into the chamber at the time of performing the film formation, the first material, the second material, and the third material become nitrides and are deposited on the surface of the base material **32**. When methane (CH_4) gas as a carbonization source is introduced into the chamber, the first material, the second material, and the third material become carbides, and are deposited on the surface of the base material **32**. When nitrogen (N_2) gas as a nitriding source and methane (CH_4) gas as a carbonization source are introduced into the chamber, the first material, the second material, and the third material become

carbonitrides, and are deposited on the surface of the base material **32**. When oxygen (O₂) gas as an oxidation source is introduced into the chamber, the first material, the second material, and the third material become oxides, and are deposited on the surface of the base material **32**.

(65) As described above, each of the first layer **L1**, the second layer **L2**, and the third layer **L3** has excellent abrasion resistance. However, the first layer **L1** has relatively lower hardness than the second layer **L2** and the third layer **L3**. In the lower layer portion **331**, such a first layer **L1** is sandwiched between the second layer **L2** and the third layer **L3**. As a result, the toughness of the lower layer portion **331** is improved. In the upper layer portion **332**, such a first layer **L1** is sandwiched between the second layers **L2**. As a result, the toughness of the upper layer portion **332** is improved. Therefore, minute cracks are less likely to occur on the surface of the coating film **33** than in a case where the coating film includes only the first layer **L1**, only the second layer **L2**, or only the third layer **L3**. As a result, the abrasion resistance of the coating film **33** is improved, and the life of the tool **30** can be prolonged.

(66) The thickness of the lamination unit **63** including the first layer **L1**, the second layer **L2**, and the third layer **L3** is desirably 200 nm or less. By suppressing the thickness of the lamination unit **63** to 200 nm or less, the lamination unit **63** can be repeatedly stacked many times in the lower layer portion **331**. Accordingly, the abrasion resistance of the coating film **33** can be further improved.

(67) In addition, the thickness of the lamination unit **61** constituted by the first layer **L1** and the second layer **L2** is also desirably 200 nm or less. By suppressing the thickness of the lamination unit **61** to 200 nm or less, the lamination unit **61** can be repeatedly stacked many times in the upper layer portion **332**. Accordingly, the abrasion resistance of the coating film **33** can be further improved.

(68) In the example of FIG. **8**, the lowermost layer of the lower layer portion **331** is the first layer **L1**, and the uppermost layer is the third layer **L3**. However, the lowermost layer of the coating film **33** may be the second layer **L2** or the third layer **L3**. The uppermost layer of the coating film **33** may be the first layer **L1** or the second layer **L2**. In addition, the stacking order of the first layer **L1**, the second layer **L2**, and the third layer **L3** in the lower layer portion **331** is not limited to the example of FIG. **8**. For example, the stacking order of the second layer **L2** and the third layer **L3** may be opposite to that in the example of FIG. **8**.

(69) In the example of FIG. **8**, the lowermost layer of the upper layer portion **332** is the first layer **L1**, and the uppermost layer is the second layer **L2**. However, the lowermost layer of the upper layer portion **332** may be the second layer **L2**. The uppermost layer of the upper layer portion **332** may be the first layer **L1**.

(70) FIG. **9** is a graph illustrating the results of an experiment for examining the abrasion resistance performance of the coating film **33**. The graph of FIG. **9** includes first data **D1** indicating the example and second data **D2** indicating the comparative example. In examples, a tool has been used in which a coating film including a lower layer portion and an upper layer portion similar to those in the fourth embodiment have been formed on the surface of the base material **32**. In the comparative example, a tool in which a conventional Al—Cr—N-based monolayer film has been formed on the surface of the same substrate as in the example has been used. The thickness of the coating film has been the same in examples and comparative examples. The type of tool has been a hob cutter.

(71) FIG. **9** is a graph illustrating the result of examining the relationship between a total cutting length *s* of a large number of parts cut using the tools of examples and comparative examples and a depth *t* of wear generated on the surface of the tool. Referring to the result of FIG. **9**, in the first data **D1** indicating the example, the depth *t* of wear generated on the surface of the tool is reduced by nearly 50% as compared with the second data **D2** indicating the comparative example. From this result, it can be seen that the abrasion resistance of the coating film of the tool of the example is significantly improved as compared with the tool of the comparative example.

(72) The coating film **33** of the tool **30** of each of the above embodiments has a lamination unit including the first layer **L1** and at least one of the second layer **L2** and the third layer **L3**. The compositions of the first layer **L1**, the second layer **L2**, and the third layer **L3** are as described above. By laminating the plurality of layers in this manner, the toughness of the coating film **33** is improved. As a result, the abrasion resistance of the coating film **33** can be improved as compared with a case where a coating film having the same thickness is formed only with one of the first layer **L1**, the second layer **L2**, and the third layer **L3**.

(73) For example, in a case where the coating film **33** is formed only of the second layer **L2**, when an extremely small crack occurs in the second layer **L2**, the crack may progress in the thickness direction of the coating film **33** to become a large crack. On the other hand, for example, when the coating film **33** is formed of the first layer **L1** and the second layer **L2**, a plurality of layers having different hardness is stacked. In this case, since the hardness of the first layer **L1** is lower than the hardness of the second layer **L2**, it is possible to suppress an extremely small crack generated in the second layer **L2** from proceeding in the thickness direction of the coating film **33** beyond the boundary between the second layer **L2** and the first layer **L1**. Therefore, large cracks in the thickness direction are less likely to occur in the coating film **33**. That is, as compared with the case where the coating film **33** is formed only of the second layer **L2**, the possibility of generating a crack larger than the thickness of the second layer **L2** can be reduced. As a result, the abrasion resistance of the coating film **33** can be improved.

(74) In particular, the coating film **33** of each of the above embodiments is a multilayer film in which a plurality of lamination units including the first layer **L1** and at least one of the second layer **L2** and the third layer **L3** are stacked. As described above, by repeatedly laminating the lamination unit a plurality of times, the abrasion resistance of the coating film **33** can be improved as compared with the case where the lamination is not repeated.

(75) In the coating film **33** of each of the above embodiments, the thickness h of the lamination unit is 200 nm or less. As described above, by suppressing the thickness of one lamination unit, the lamination unit can be repeatedly stacked in the coating film **33** many times. Accordingly, the abrasion resistance of the coating film **33** can be further improved.

(76) As described above, each of the first layer **L1**, the second layer **L2**, and the third layer **L3** may be any of a nitride, a carbide, a carbonitride, and an oxide. However, if the first layer **L1** is a nitride of the first material, the second layer **L2** is a nitride of the second material, and the third layer **L3** is a nitride of the third material, the abrasion resistance of the coating film **33** can be further improved as compared with the case of carbide, carbonitride, or oxide.

(77) The tool **30** includes the base material **32** and the coating film **33** formed on the surface of the base material **32**. As described above, the coating film **33** has extremely high abrasion resistance by having a lamination unit including the first layer **L1** and at least one of the second layer **L2** and the third layer **L3**. As a result, the life of the tool **30** can be improved as compared with the case of having a single-coating film coating formed of a single material.

(78) While an exemplary embodiment of the present disclosure has been described above, the present disclosure is not limited to the above embodiment.

(79) In the above-described embodiment, the above-described lamination unit is directly deposited on the surface of the base material **32**. However, the above-described lamination unit may be formed on the surface of the base material **32** via another layer. For example, a single layer portion including only the third layer **L3** may be formed on the surface of the base material **32**, and the above-described lamination unit may be stacked on the surface of the single layer portion.

(80) Further, in the above embodiment, the case where the tool **30** is a hob cutter for cutting the external teeth of the gear has been described. However, the application target of the coating film **33** is not limited to the hob cutter. FIG. **10** is a perspective view of a skiving cutter as another example of the tool **30**. The skiving cutter is a tool **30** for cutting internal teeth of a gear. The coating film **33** similar to that in each of the above embodiments may be formed on the surface of such a skiving

cutter. The tool **30** to which the coating film **33** is applied may be a pinion cutter, a shaving cutter, or the like.

(81) It is desirable that the tool **30** be any one of the above-described hob cutter, pinion cutter, skiving cutter, and shaving cutter. Since these cutting tools cut a metal part, the coating film on the surface is required to have very high hardness. However, since a material having high hardness has low toughness, there is a problem that cracks are likely to occur. In general, it is difficult to achieve both hardness and toughness with a single material.

(82) In this respect, the coating film **33** of each of the above embodiments is formed by laminating at least one of the second layer **L2** and the third layer **L3** having high hardness and the first layer **L1** having hardness relatively lower than that of the second layer **L2** and the third layer **L3**. As a result, a buffering action by the first layer **L1** is obtained, and the toughness of the coating film **33** is improved. Therefore, cracks are less likely to occur in the coating film **33** than when the coating film is formed of a single material. That is, by forming the coating film **33** into a multilayer structure of a plurality of materials, it is possible to achieve both hardness and toughness required for a tool. In addition, even if a minute crack occurs in the second layer **L2** or the third layer **L3** in the vicinity of the surface of the coating film **33**, the first layer **L1** under the second layer **L2** or the third layer **L3** has high toughness, so that the crack hardly proceeds in the depth direction. As a result, the abrasion resistance of the coating film **33** is improved, and the life of the tool **30** can be prolonged.

(83) The present application can be applied to, for example, a coating film, a tool, and a machine tool.

(84) Features of the above-described preferred embodiments and the modifications thereof may be combined appropriately as long as no conflict arises.

(85) While preferred embodiments of the present disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present disclosure. The scope of the present disclosure, therefore, is to be determined solely by the following claims.

Claims

1. A coating film, comprising a lamination unit including a first layer and at least one of a second layer and a third layer, wherein the first layer contains a nitride, a carbide, a carbonitride, or an oxide of a first material of which elemental composition is represented by $(\text{Cr}_{\text{sub.1-a-b-cAl}_{\text{sub.a}}[\text{Ni}_{\text{sub.1-dZr}_{\text{sub.d}}]_{\text{sub.bX}_{\text{sub.c}}})$, as features of the first layer, the $[\text{Ni}_{\text{sub.1-dZr}_{\text{sub.d}}}]$ is a ZrNi compound, the X is at least one element selected from Ti, Nb, Si, B, W, and V, the a, b, and c respectively represent atomic concentrations of Al, [NiZr], and X in the $(\text{Cr}_{\text{sub.1-a-b-cAl}_{\text{sub.a}}[\text{Ni}_{\text{sub.1-dZr}_{\text{sub.d}}]_{\text{sub.bX}_{\text{sub.c}}})$, and satisfy $0.5 \leq a \leq 0.8$, $0.01 \leq b \leq 0.35$, and $0 < c \leq 0.2$, the d represents an atomic concentration of Zr in the $[\text{Ni}_{\text{sub.1-dZr}_{\text{sub.d}}}]$, and satisfies $0.2 \leq d \leq 0.5$, the second layer contains a nitride, a carbide, a carbonitride, or an oxide of a second material of which elemental composition is represented by $(\text{Al}_{\text{sub.e}}\text{Cr}_{\text{sub.1-e-fZ}_{\text{sub.f}}})$, the Z is at least one element selected from Si, Y, and B, the e and f represent atom concentrations of Al and Z in the $(\text{Al}_{\text{sub.e}}\text{Cr}_{\text{sub.1-e-fZ}_{\text{sub.f}}})$, respectively, and satisfy $0.5 \leq e \leq 0.8$, $0.03 \leq f \leq 0.3$, $e+f \leq 0.9$, the third layer contains a nitride, a carbide, a carbonitride, or an oxide of a third material of which elemental composition is represented by $(\text{Al}_{\text{sub.g}}\text{Cr}_{\text{sub.1-g}})$, and the g represents an atomic concentration of Al in the $(\text{Al}_{\text{sub.g}}\text{Cr}_{\text{sub.1-g}})$, and satisfies $0.5 \leq g \leq 0.8$.
2. The coating film according to claim 1, wherein a plurality of the laminating units are stacked.
3. The coating film according to claim 2, wherein a thickness of the lamination unit is 200 nm or less.
4. The coating film according to claim 2, wherein a plurality of lamination units including the first layer and the second layer are stacked.

5. The coating film according to claim 2, wherein a plurality of lamination units including the first layer and the third layer are stacked.
 6. The coating film according to claim 2, wherein a plurality of lamination units including the first layer, the second layer, and the third layer are stacked.
 7. The coating film according to claim 2, comprising: a lower layer portion in which a plurality of lamination units including the first layer, the second layer, and the third layer are stacked; and an upper layer portion in which a plurality of lamination units including the first layer and the second layer are stacked.
 8. The coating film according to claim 1, wherein the Z is Si or Y.
 9. The coating film according to claim 1, wherein the first layer is the nitride of the first material, the second layer is the nitride of the second material, and the third layer is the nitride of the third material.
 10. A tool, comprising: a base material; and the coating film according to claim 1 formed on a surface of the base material.
 11. The tool according to claim 10, wherein the tool is any one of a hob cutter, a pinion cutter, a skiving cutter, and a shaving cutter.
 12. A machine tool, comprising: a part holder; and the tool according to claim 11 that machines a part held by the part holder.
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